MILITARY SPECIFICATION

CARTRIDGE, 20MM, ARMOR PIERCING-TRACER, M95 (PERCUSSION PRIMER)

This specification is approved for use within the US Army Armament Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers requirements, quality assurance provisions and preparation for delivery for Cartridge, 20mm, Armor Piercing Tracer, (APT), M95 for use in the M3 and MK16 20mm Automatic Weapon Systems.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, and standards. The following specifications, and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issue of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification for

STANDARDS

MILITARY

MIL-STD-105 -Sampling Procedures and Tables for Inspection by Attributes

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Armament, Munitions and Chemical Command, Attn. AMSMC-QA, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1305
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MIL-STD-644 - Visual Inspection Standards and Inspection Procedures for Inspection of Packaging, Packing and Marking for Small Arms Ammunition
MIL-STD-651 - Visual Inspection Standards for 20mm Ammunition and Components

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS (See 6.5)

US ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

PRODUCT AND PACKAGING DRAWINGS

12576573 - Cartridge, Armor Piercing Tracer, M95
12576575 - Charged Projectile, 20mm, AP-T for the M95 Cartridge
12576576 - Closing Cup
7553663 - Classification of Firing Casualties for: Case, Cartridge, 20mm, M21A1 and M21A1B1

INSPECTION EQUIPMENT DRAWINGS

7259443 - Rack Cartridge
7259469 - Torque Test Gage
7286068 - Limit Check Taper Plug Gage
7286069 - Limit Check Taper Plug Gage
7286070 - Limit Check Plug Gage
7276071 - Limit Check Plug Gage
7299942 - Gage, Chamber
7556901 - Indicator Comparator Gage
11075131 - Fixture, Test

PUBLICATIONS

US. ARMY MATERIAL COMMAND

AMCR 715-505- Ammuniton Ballistic Acceptance Test Methods, Volume 8, Test Procedure for 20mm Cartridges

(Copies of specifications, standards, handbooks drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

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2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3 REQUIREMENTS

3.1 General.

a. Material. Materials shall be in accordance with the applicable drawings and specifications.

b. Manufacturing process. All parts and assemblies shall be manufactured by a process approved by the contracting officer, and no deviation from that process shall be made without prior approval (6.6).

3.2 Cartridge. The cartridge shall comply with all the requirements specified on Drawing (Dwg.) 12576573, all associated drawings and with all requirements specified in applicable specifications,

a. Projectile extraction. The force necessary to extract the projectile from the cartridge case shall be 1100 to 2800 pounds, inclusive.

b. Projectile torque. The projectile shall withstand a minimum torque of 10 inch pounds without movement.

3.3 Contamination of energetic components. Extreme care shall be exercised to avoid contamination of primers and propellant by oil, grease or other foreign matter since such contamination can cause misfires, increased action time and lot rejection.

3.4 Functioning. The cartridge shall comply with the following:

3.4.1 Conditioned at 68°F to 72°F.

a. Velocity. The average velocity of the projectile shall be 2680 ± 50 feet per second (fps) at 78 feet from the muzzle of the gun and the standard deviation of the velocity shall not exceed 40 fps. (see 6.7).

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b. Pressure. The average chamber pressure ($\bar{X}$) plus 0.84 times the average range (R) (see 6.8) of the chamber pressure shall not exceed 51,000 pounds per square inch (psi).

c. Waterproofness. The average velocity (wet) of projectiles shall not vary from the average velocity (dry) of projectiles by more that 100 fps.

3.4.2 Conditioned at ambient temperature.

a. Function and casualty. Cartridges shall function without casualty when fired in the M3 or MK16 20mm automatic weapon systems.

b. Accuracy. The average mean radius of all targets shall not exceed 15 inches (accuracy) at a range of 600 yards.

c. Projectile trace. The projectile shall exhibit a trace visible against a dark background for a minimum of 2.0 seconds.

3.4.3 Conditioned at -65°F.

a. Action time. The action time of the sample cartridges and the average action time ($\bar{X}$) plus four sample standard deviations shall not exceed 4.0 milliseconds (see 6.7).

b. Projectile trace. The projectile shall exhibit a trace visible against a dark background for a minimum of 2.0 seconds.

3.4.4 Conditioned at +145°F.

a. Projectile trace. The projectile shall exhibit a trace visible against a dark background for a minimum of 2 seconds.

3.4.5 Safety. The projectile shall not stick in the gun bore in any of the test firings.

3.5 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection in accordance with the technical provisions herein (see 4.3 and 6.13).

3.6 Workmanship. The requirements for workmanship are as specified by applicable drawings, referenced specifications and the following:

3.6.1 Processing defects. The metal parts and assemblies shall be free of splits, perforations, scratches (other than draw scratches), folds wrinkles, buckles, bulges, dents, cracks, rupture, deformaties and other forms of substandard workmanship.

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3.6.2 Cleanliness. All components and assemblies shall be free of foreign matter, such as oil, grease, corrosion, dirt, chips, etc.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may utilize his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system of quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of any defective material.

4.2 Classification of inspections. The following types of inspection shall be conducted on this item.

a. First Article Inspection (see 4.3)

b. Quality Conformance Inspection (see 4.4)

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated.

Parts description	Drawing	Quantity
Closing, Cup	12576576	125
Charged Projectile	12576575	125
Cartridge	12576573	480
Packaging (containers)	(see 5.1)	7

4.3.2 Inspections to be performed. See MIL-A-48078 and Table I specified herein.

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4.3.3 Rejection. See MIL-A-48078. Failure of the original first article sample to be in accordance with the applicable drawings and requirements of Table I, as specified herein will result in sample disapproval.

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TABLE I. First article inspection

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-70636 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER
	Cartridge, 20mm, APT, M95 and components 1/				See Below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Closing, Cup - (Dwg. 12576576)				
	Examination for defects	125	Acc-0 Rej-1	3.2	4.4.2.1
	Charged Projectile - (Dwg. 12576575)				
	Examination for defects	125	Acc-0 Rej-1	3.2	4.4.2.2
	Cartridge - (Dwg. 12576573)				
	Examination for defects	480	Acc-0 Rej-1	3.2	4.4.2.3
	Test Requirements				
	Non Functioning				4.5.1
	(1) Projectile extraction	20	Acc-0 Rej-1	3.2a	4.5.1.1/11075131/ 7259443 2/
	(2) Projectile torque	20	Acc-0 Rej-1	3.2b	4.5.1.2/7259469
	(3) Propellant contamination/ Charge weight verification	20	Acc-0 Rej-1	3.3	4.5.1.3 2/

NOTES: 1/ MIL-STD-651 and Dwg. 7553663 shall apply in defining and evaluating cartridge visual defects. 2/ Test may be conducted using same sample cartridges. 3/ Failure of the cartridges to comply with the applicable requirements shall be cause for rejection of the First article quantity. Second sample not permitted.

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TABLE I. First article inspection

CLASSIFICATION OF DEFECTS & TESTS MIL-C-70636 (AR)

PARAGRAPH	TITLE: Cartridge, 20mm, APT, M95 and components 1/		SHEET 2 OF 2	DRAWING NUMBER / NEXT HIGHER ASSEMBLY	
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Functioning			3.4	
	(1) Conditioned at 68°F to 72°F			3.4.1	4.5.2.1
	a. Velocity	20	3/	3.4.1a	4.5.2.2 4/
	b. Pressure	20	3/	3.4.1b	4.5.2.2 4/
	c. Waterproofness	20	3/	3.4.1c	4.5.2.2
	(2) Conditioned at ambient			3.4.2	4.5.3
	a. Function and casualty	200	Acc-0 Rej-1	3.4.2a	4.5.3.1
	b. Accuracy	50	3/	3.4.2b	4.5.3.2
	(3) Conditioned at -65°F			3.4.3	4.5.4.1
	a. Action time	50	3/	3.4.3a	4.5.4.2
	b. Projectile trace	50	Acc-0 Rej-1	3.4.3b	4.5.5.2
	(4) Conditioned at +145°F			3.4.4	4.5.5.1
	a. Projectile trace	50	Acc-0 Rej-1	3.4.4a	4.5.5.2
	Packaging (containers)				
	Examination for defects	7	Acc-0 Rej-1	3.1a	4.4.4/5.1

NOTES: 4/ Velocity and pressure readings shall be recorded in a single test using the same sample cartridges.

AMSMC Form 1570, 1 Feb 85 Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with lot formation provisions of MIL-A-48078. In addition, each inspection lot shall contain.

a. Cartridge cases from one interfix lot number from one manufacturer.

b. Projectiles from one interfix lot number from one manufacturer.

c. Primers from one interfix lot number from one manufacturer and no more than three lot numbers.

d. Propellant from one lot number from one manufacturer.

4.4.2 Examination. See MIL-A-48078.

a. Unless otherwise specified in the Classification of Defects and Test Tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

b. Major and minor defects. To determine product acceptability, major and minor defects as listed herein may be considered collectively on a class basis or defects may be considered individually. When a class basis is utilized the Acceptable Quality Level (AQL) for major defects shall be 1.5 percent and the AQL for minor defects shall be 2.5 percent. However, when major defects are considered collectively on a class basis, the AQL for any individual defect shall be limited to .40 percent. In addition, when three or less defects appear in a classification, sampling plans applicable to a class shall not be used for major defects. (See 6.9).

c. Critical defects. Unless otherwise specified, one hundred percent examination shall be performed for all critical defects. If a visual critical defect is found in a sample either just prior to a firing test or after a firing test (and the defect is not due to the firing), the lot shall be rejected. The lot shall then be screened and resubmitted for visual inspection of critical defects. If a critical defect is found during packing, the portion of the lot that has been packed or is in the process of being packed shall be rejected. In addition the portion of the lot remaining to be packed shall be rejected. The lot shall then be screened and resubmitted for visual inspection of critical defects.

d. Cartridge. Each cartridge shall be inspected in process for chamber gage failure (profile and alignment, max.). Cartridges which fail to meet the requirement of 3.2 shall be removed.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-70636 (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.1	Closing, Cup, Proj, 20mm, APT, M95		12576576
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined				
MAJOR					
101.	.0025 Thickness cup base		.25	3.2	Gage
102.	.308 Diameter		.25	3.2	Gage
103.	Punctured cup		.25	3.2	Visual
104.	.130 Height cup		.25	3.2	Gage
MINOR					
201.	Scoring missing or incomplete		.40	3.2	Visual
202.	Poor workmanship		.65	3.6	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-70636 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.2	Charged Projectile, 20mm, AP-T, for the M95 Cartridge				12576575
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined				
MAJOR					
101.	1.40 Depth, base to tracer mix		.25	3.2	Gage
102.	Cup closure missing or perforated		.25	3.2	Visual
103.	Sealant missing around cup		.25	3.2	Visual
104.	3.270 Overall length, max		.25	3.2	Gage
MINOR					
201.	Poor workmanship		.65	3.6	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-70636 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.3	Cartridge, 20mm, APT, M95 1/ (examinations)	1 OF 2	12576573
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1.	Split or perforated case		100%	3.2	Visual
2.	Crack or split in projectile		100%	3.2	Visual
3.	Case to projectile crimp missing		100%	3.2	Visual
Major					
101.	Chamber gage failure (profile & alignment, max)	300	Acc-0 Rej-1	3.2	7299942/7286068/ 2/ 7286069/7286070/ 7286071
102.	Depth of primer seating		.25	3.2	Gage
103.	Total length, max		.25	3.2	7556901 3/
104.	Primer cocked, loose, missing or inverted		.25	3.2	Manual
105.	Corrosion		.25	3.2	Visual
106.	Head Configuration		.25	3.2	Visual
107.	Improper or incomplete crimp (case-projectile)		.25	3.2	Visual
108.	Wax coating missing or incomplete		.25	3.2	Visual

NOTES:
1/ MIL-STD-651 and 7553663 shall apply in defining and evaluating cartridge visual defects. 2/ A dead weight load of 60 pounds maximum may be used to insert the cartridge in the gage. 3/ Setting check block 7.2300-.0004.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-70636 (AR)

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER
4.4.2.3	Cartridge, 20mm, APT, M95 1/ (examinations)				12576573
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Minor					
201.	Upset (crooked) or deformed windshield		.40	3.2	Visual
202.	Waterproofing missing around primer		.40	3.2	Visual
203.	Primer crimp missing or incomplete		.40	3.2	Visual
204.	Rotating band damaged		.40	3.2	Visual
205.	Foreign matter except corrosion		.40	3.2	Visual
206.	Projectile protective coating damaged or incomplete		.40	3.2	Visual
207.	Marking incorrect, incomplete, illegible or missing		.40	3.2	Visual
208.	Cartridge Case damaged		.40	3.2	Visual
209.	Scratches (other than draw scratches), folds, wrinkles, bulges, dents		.40	3.2	Visual
210.	Wax coating on projectile		.40	3.2	Visual
211.	Wax coating flaky or otherwise unsatisfactory		.40	3.2	Visual
212.	Poor workmanship		.65	3.6	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-70636 (AR)

PARAGRAPH	TITLE			SHEET 1 OF 2	DRAWING NUMBER
4.4.2.4	Cartridge, 20mm, APT, M95 (Tests) 1/				12576573
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Tests				
	Non-functioning				
	(1) Projectile extraction	20	4.4.3.1	3.2.a	4.5.1.1/11075131/ 7259443/ 2/
	(2) Projectile torque	20	4.4.3.2	3.2.b	4.5.1.2/7259469
	(3) Propellant contamination/Charge weight verification	20	Acc-0 Rej-1	3.3	4.5.1.3 2/
	Functioning			3.4	
	(1) Conditioned at 68°F to 72°F		4.4.3.4	3.4.1	4.5.2/4.5.2.1
	a. Velocity	20	4.4.3.4.a	3.4.1.a	4.5.2.2 3/
	b. Pressure	20	4.4.3.4.b	3.4.1.b	4.5.2.2 3/
	c. Waterproofness	20	4.4.3.4.c	3.4.1.c	4.5.2.2
	(2) Conditioned at Ambient			3.4.2	4.5.3
	a. Function & Casualty	200	4.4.3.5	3.4.2.a	4.5.3.1
	b. Accuracy	50	4.4.3.6	3.4.2.b	4.5.3.2
	c. Projectile trace	50	4.4.3.7	3.4.2.c	4.5.5.2
	(3) Conditioned at -65°F			3.4.3	4.5.4/4.5.4.1
	a. Action time	50	4/	3.4.3a	4.5.4.2

NOTES:

AMSMC Form 1570, 1 Feb 85 Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-70636 (AR)

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER
4.4.2.4	Cartridge, 20mm, APT, M95 (Tests) 1/				12576573
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	1/ MIL-STD-651 and Dwg. 7553663 shall apply in defining and evaluating cartridge visual defects. 2/ Test may be conducted using same sample cartridges. 3/ Velocity and pressure readings shall be recorded in a single test using the same sample cartridges. 4/ Failure of the cartridges to comply with the applicable requirement shall be cause for rejection of the lot.				

NOTES:

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4.4.3 Tests. The test listed in 4.4.2.4 shall be performed on each cartridge lot in accordance with the test methods prescribed in 4.5. Sample size and acceptance criteria for each test shall be as specified. Only cartridges which have met the visual and dimensional requirements and which have been selected in such a manner that the sample is representative of the entire lot shall be used in the tests. The selected cartridges shall be mixed thoroughly before being divided into samples for the various tests.

4.4.3.1 Projectile extraction.

a. Failure of two or more sample cartridges to comply with the minimum requirement shall cause rejection of the lot. If one cartridge of the sample fails to comply with the minimum requirement, a second sample of 20 cartridges shall be tested. The lot shall be rejected if two or more cartridges of the combined sample fail to comply with the minimum requirement.

b. Failure of four or more cartridges to comply with the maximum requirement shall be cause for rejection of the lot. If more than one but less than four cartridges fail to comply with the maximum requirement, a second sample of 20 cartridges shall be tested. The lot shall be rejected if five or more cartridges of the combined sample fail to comply with the maximum requirement.

4.4.3.2 Projectile torque. Failure of two of more of the sample cartridges to comply with the specified torque requirement shall cause rejection of the lot. If one cartridge of the sample fails to comply, a second sample of 20 cartridges shall be tested. The lot shall be rejected if two or more cartridges of the combined samples fail to comply with the applicable requirement.

4.4.3.3 All cartridge firing tests. The lot shall be rejected if in any firing test one or more of the following casualties occur:

a. Misfire associated with c or d of 6.10.

b. Projectile remaining in bore.

c. Metal parts separation, except rotating band separation (see 6.11.1).

4.4.3.4 Functioning at 68°F to 72°F. If any requirement(s) in 3.4.1 is not achieved on the first sample, a second sample shall be tested for the failed parameter(s).

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a. Velocity. If the sample fails to comply with either or both requirements a retest sample of 40 cartridges shall be tested. The lot shall be rejected if the retest samples fails to comply with the applicable requirement(s). (See 6.7)

b. Pressure. If the average pressure ($\bar{X}$) plus 0.84 times the average range ($\bar{R}$) of the sample cartridges exceeds 51,000 psi a retest sample of 40 cartridges shall be tested. The lot shall be rejected if the average pressure ($\bar{X}$) plus 0.91 times the average range ($\bar{R}$) of the retest sample fails to comply with the applicable requirements. (See 6.8).

c. Waterproofness. If the average velocity (wet) of sample fails to comply with the applicable requirement, a retest sample of 40 cartridges shall be tested. The lot shall be rejected if the average velocity (wet) differs by more than 100 fps from the average velocity (dry) of 40 additional sample cartridges fired to establish a new base for comparison. The lot shall not be penalized for failure of the last named sample cartridges to comply with the requirements of 3.4.1.a.

4.4.3.5 Function and casualty. Two-hundred cartridges shall be subjected to this test. The lot shall be rejected if, at any time, any of the following occurs:

a. Any fire "out of battery" (see 6.11.2).

b. Any primer misfires (see 6.10).

c. Any cartridge case fails to extract.

d. Any cartridge case longintudinal split occurs (L or M) (Dwg. 7553663).

e. An cartridge case circumferential rupture, partial (L) or complete (Dwg. 7553663).

f. Any detached metal from cartridge case occurs (see 6.11.3).

g. Any cartridge fails to chamber.

h. Projectile remaining in bore.

In addition, if defects listed in Table II, occur equal to or less than the "ACC" number, the lot shall be accepted. If any defect(s) occur equal to or exceeding the "REJ" number, the lot shall be rejected. If any defect(s) occur in excess of the "ACC" number, but less than the "REJ" number, a second sample of 400 cartridges shall be selected and fired in the M3 or MK 16 weapons. The lot shall be rejected if any defect occurs equal to or greater than the "CUMMULATIVE REJ" number.

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TABLE II. Firing defects

		ACC	REJ	CUMULATIVE REJ
1.	Primer missing	0	2	2
2.	Primer leak (see Note 1)	9	20	25
3.	Longitudinal case split (I or S) (Dwg. 7553663)	10	21	31
4.	Longitudinal case split (K) (Dwg. 7553663)	0	2	2
5.	Case circumferential rupture, partial (S, J or K) (Dwg. 7553663)	4	9	11
6.	Longitudinal case split (J) (Dwg. 7553663)	4	9	11
7.	Projectile rotating band separation complete or partial separation having band seat serration imprint (see 6.11.4)	0	3	3
8.	Projectile rotating band separation having greater than 0.5 inch and without band seat serration imprint (see 6.11.4)	1	4	6

Note 1. For primer defect definitions, see Chapter 1, Section 3, Paragraph 3.6 of AMCR 715-505, Volume 8.

4.4.3.6 Accuracy. If the average mean radius of the sample cartridges exceeds the applicable requirement, a retest sample of 100 cartridges shall be tested. The lot shall be rejected if the average mean radius of 10 retest accuracy targets exceeds the applicable requirement (see 3.4.2.b).

4.4.3.7 Projectile trace. Failure of 7 or more sample cartridges fail to comply with trace requirement shall be cause for rejection of the lot. If more than three but less than 7 cartridges fail, a second sample of 50 shall be tested. The lot shall be rejected if nine (9) or more cartridges of the combined samples fail to meet the requirement.

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4.4.4 Packaging, packing and marking. Inspection for packaging, packing and marking shall be in accordance with MIL-STD-644.

4.4.5 Inspection equipment. The inspection equipment required to perform the examination and tests prescribed herein is described in the "Paragraph Reference/Inspection Method" column in the tables starting with paragraph 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6.12 herein.

4.4.5.1 Ballistic test equipment. The ballistic test equipment required to perform the test prescribed herein shall be as specified in AMCR 715-505 and by the design activity.

4.5 Methods of inspection.

4.5.1 Non-functioning.

4.5.1.1 Projectile extraction. The cartridge shall be placed in the applicable fixture specified herein and the tension force specified on the applicable drawing shall be applied. The projectile must separate from the cartridge case in the range specified on the applicable drawing. The method of test shall be as specified in Chapter 1, Section 10 of AMCR 715-505, Volume 8. This is a destructive test.

4.5.1.2 Projectile torque. The cartridge shall be marked with a light scratch extending axially across the rotating band onto the case neck. Then the cartridge shall be placed in the applicable fixture specified herein and the inch pound torque specified on the applicable drawing shall be applied slowly to the projectile. Observation shall be made for movement of the projectile with respect to the misalignment in the scratch mark. This test is a non-destructive test.

4.5.1.3 Propellant contamination and charge weight verification. After completion of test specified in 4.5.1.1 each of the cartridges tested shall be examined visually for contamination as described in Chapter 1, Section 10 of AMCR 715-505, Volume 8. The charge from each of the sample cartridges shall be weighed and recorded.

4.5.2 Functioning (conditioned at 68°F to 72°F).

4.5.2.1 Ammunition conditioning. The cartridges shall be conditioned for a period of not less than six (6) hours after the conditioning chamber has stabilized at 68°F to 72°F. Timing for the conditioning period shall start when the chamber has stabilized following the sample being placed in the conditioning chamber.

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4.5.2.2 Velocity, pressure and waterproofness. These tests shall be conducted in accordance with Chapter 1, Section 2 of AMCR 715-505, Volume 8. The sample standard deviation shall be computed in accordance with 6.7 and 6.8 respectively.

4.5.3 Functioning (conditioned at ambient temperature).

4.5.3.1 Function and casualty. The method of testing shall be as specified in Chapter 1, Section 3, of AMCR 715-505, Volume 8. The test sample shall be fired in bursts of 50 cartridges in the M3 or MK 16 gun. The weapon barrel shall be at ambient temperature at the beginning of test and cooled to ambient after each 100 rounds.

4.5.3.2 Accuracy. The method of test and measurement of targets shall be in accordance with Chapter 1, Section 4, of AMCR 715-505, Volume 8. The specified range may be shortened in decrements of 100 yards, to a minimum range of 200 yards. For ranges less than 600 yards, the average mean radius shall be reduced in direct proportion to the reduction in range. Any variation from the 200 yard range requires direct proportional adjustment of the average mean radius.

4.5.4 Functioning (conditioned at -65°F).

4.5.4.1 Ammunition conditioning. The cartridges shall be conditioned for a period of not less than six (6) hours after the conditioning chamber has stabilized at -65°F $\pm$ 5°F. Timing for the conditioning period shall start when the chamber has stabilized following the sample being placed in the conditioning chamber.

4.5.4.2 Action time. The method of testing shall be in accordance with Chapter 1, Section 5, of AMCR 715-505, Volume 8.

4.5.5 Funtioning (conditioned at +145°F).

4.5.5.1 Ammunition conditioning. The cartridges shall be conditioned for a period of not less than six (6) hours after the conditioning chamber has stabilized at +145°F $\pm$ 5°F. Timing for the conditioning period shall start when the chamber has stabilized following the sample being placed in the conditioning chamber.

4.5.5.2 Projectile trace. The method of test shall be as specified in Chapter 1, Section 8, of AMCR 715-505, Volume 8. Projectile trace time shall begin at cartridge ignition and continue until end of the tracer burn. Observation shall be from near the gun position.

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4.5.6 Defective weapon. If a test weapon fails to comply with the minimum cyclic rate during a normal burst operation or its found to be defective during or at the end of burst, the test shall be suspended and Government approval shall be required for a new test. The cyclic rate of each burst shall not be less than 680 shots per minute in the M3 or MK16 gun. The cyclic rate for each burst shall be recorded.

4.5.7 Test validity. If for any reason the test activity considers that the test conditions have detrimentally affected the test results, the test activity may request the Government to declare the test invalid and authorize a new test.

5. PACKAGING

5.1 Preservation and packaging. The M95 cartridge shall be preserved and packaged in accordance with instructions and the applicable Combination of Adopted Items Drawing from the procuring agency.

5.2 Packing. The cartridge shall be packed in accordance with the Combination of Adopted Items Drawing from the procuring agency.

5.3 Marking. Marking of the packed container shall be in accordance with the applicable Combination of Adopted Items Drawing.

5.4 Shipping. When shipments from more than one lot are shipped as a carload, each lot shall be kept separate and the division between lots clearly indicated to prevent mixing of lots in transit.

6. NOTES

6.1 Intended use. This specification covers the loading assembling and packing for Cartridge, 20mm, Armor Piercing Tracer M95 for use in M3 or MK16 weapons designed to utilize percussion primed cartridges with the M21 type case.

6.2 Ordering data. See MIL-A-48078.

6.3 Submission of test data. In addition to the normal distribution of records, when the cartridge is procured by AMCCOM, one (1) copy of all ballistic data and ammunition data cards shall be forwarded to: Commander, ARDEC, ATTN: AMSMC-QAF-S(D) and SMCAR-CCL-CE, Picatinny Arsenal, NJ 07806-5000.

6.4 Safety data. For the safety data requirements refer to Safety Data Document as listed on Product Drawings.

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6.5 Drawings. Drawings listed in Section 2 of this specification under the heading U.S. Army Armament Research, Development and Engineering Center (ARDEC) may also include drawings prepared by and identified as Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal, Picatinny Arsenal, U.S. Army Armament Research and Development Command (ARRADCOM) or U.S. Army Armament Reseach and Development Center (ARDC) drawings. Technical data originally prepared by these activities are now under the cognizance of ARDEC.

6.6 Process deviation. A process deviation is defined as a change in the approved basic method of manufacture, or an operational change which may alter the metallurgical or physical properties of the item.

6.7 Standard deviation. Standard deviation shall be calculated from the following formula:

$$S = \sqrt{\Sigma(X_i - \bar{X})^2/(n-1)}$$

where:

X_i = each individual value
$\bar{X}$ = sample arithmetic mean $= \Sigma X_i/n$
n = sample size
$\Sigma(X_i - \bar{X})^2$ = The sum of the squares of the differences between the sample mean and each individual value

6.8 Range plan. Where a range plan is specified, the individual values shall be recorded in subgroups of 5 in the order in which they are observed. The ranges for the subgroups will be added and the sum of the ranges shall be divided by the number of subgroups to determine the average range ($\bar{R}$).

6.9 Intermediate point inspection. The classification of defects identifies the defect characteristics for acceptance inspection. It may be necessary to modify the sequence of inspection stations to best suit the manufacturing process. Inspection for defect characteristics which will be hidden or altered by subsequent processing operations (including unrelated operations), should be scheduled to prevent premature acceptance which could be detrimental to the attainment of optimum product quality of the end item.

6.10 Misfires. Misfired test cartridges, except those from the waterproofness test shall be retested in equipment identified in the accuracy test. If a cartridge fails to function in that test, examinations shall be conducted for the following conditions:

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a. Excessive primer seating depth.

b. Missing or obstructed case vent.

c. Missing or insufficient primer composition or propellant.

d. Partially burned or contaminated primer composition or propellant.

The lot shall not be penalized if condition a. is observed. However, this condition shall be reported for information. The "ACC" number shall be zero (0) and the "REJ" number 3, for misfires associated with condition b. The second sample for evaluation of misfires associated with condition b, shall consist of the function and casualty second sample (see 4.4.3.5). The lot shall be rejected if either condition c or d is observed. Misfire(s) occurring in the waterproofness test shall be handled in accordance with AMCR 715-505, Volume 8.

6.11 Defect definitions.

6.11.1 Metal part separation. Separation or breakup of projectile part(s), except rotating band separation, as evidenced by recovery of the part(s) or fragment(s), or hole(s) in the fragmentation screen(s).

6.11.2 "Out-of-battery". A cartridge shall be classed defective if it fires, due to delayed ignition, after the bolt has been unlocked and extraction has been initiated or completed.

6.11.3 Detached metal from cartridge case. Metal sheared or missing from the fired cartridge case exterior, such as rim or neck shears. Shavings of metal from the interior wall of the case neck, in the crimped area shall not be considered as detached metal.

6.11.4 Rotating band separation. For classification as a defect, there must be evidence by recovery of the band, or portion thereof, or by hole(s) in the fragmentation screen(s). Normal band fringing shall not be classed as a defect.

6.12 Submission of inspection equipment design for approval. See MIL-A-48078. Submit equipment designs as required to Commander, ARDEC, ATTN: AMSMC-QAF-I(D), Picatinny Arsenal, NJ 07806-5000. This address will be specified on the contract Data Requirements List, DD From 1423 in the contract. Unless otherwise specified, data item DI-R-1714 will apply.

6.13 First article sample. In order to minimize inspection costs, the firing tests of 4.4 shall be performed after the sample has been provisionally accepted by the Government for all other requirements. Additional cartridges may be required by the test facility.

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6.14 Subject term (key words) listing.

Armor Piercing Tracer
Small Arms Ammunition
MK16 20mm Automatic Weapon System

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1305-AC72)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-70636	Cartridge, 20MM, Armor Piercing-Tracer, M95

3a. NAME OF SUBMITTING ORGANIZATION

b. ADDRESS *(Street, City, State, ZIP Code)*

4. TYPE OF ORGANIZATION *(Mark one)*

- ☐ VENDOR
- ☐ USER
- ☐ MANUFACTURER
- ☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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